THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 1: **Real Numbers** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The prime factorisation of 5005 is: 1

(a) $5 \times 7 \times 11 \times 13$ **(b)** $5 \times 7 \times 13$ **(c)** $5 \times 11 \times 13$ **(d)** $7 \times 11 \times 13$

2. The LCM of 12, 15 and 21 is: 1

(a) 3 **(b)** 210 **(c)** 420 **(d)** 1260

3. The HCF of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$ is: 1

(a) 36 **(b)** 180 **(c)** $2^3 \times 3^3$ **(d)** 7

4. If $a = x^3y^2$ and $b = xy^3$, where x, y are primes, then HCF(a, b) is: 1

(a) xy **(b)** xy^2 **(c)** x^3y^3 **(d)** x^2y^2

5. A student states: "The sum of a rational and an irrational number is rational." The correct response is: 1

(a) True **(b)** False; e.g. $2 + \sqrt{3}$ is irrational
(c) True only for integers **(d)** Cannot be determined

6. Which of the following is rational? 1

(a) $\sqrt{12}$ **(b)** π **(c)** $(2 + \sqrt{3})(2 - \sqrt{3})$ **(d)** $3\sqrt{5}$

7. Two students find the LCM of 8 and 9. Student A says 72; Student B says 1 because the numbers are co-prime. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. The exponent of 2 in the prime factorisation of 144 is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

9. If HCF$(26, 169) = 13$, then LCM$(26, 169)$ is: 1

(a) 338 **(b)** 13 **(c)** 26 **(d)** 169

10. The product of a non-zero rational number and an irrational number is: 1

(a) always rational **(b)** always irrational
(c) sometimes rational **(d)** always an integer

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the HCF and LCM of 6, 72 and 120 using prime factorisation. 2

12. Explain why $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite number. 2

13. Check whether 6^n can end with the digit 0 for any natural number n. 2

14. Two numbers have HCF 16 and product 3072. Find their LCM. Can two numbers have HCF 16 and LCM 380? Give reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{5}$ is irrational. 3

16. Find the largest number which divides 245 and 1029 leaving remainder 5 in each case. 3

17. Prove that $6 + \sqrt{2}$ is irrational. 3

OR

Prove that $3\sqrt{2}$ is irrational. 3

18. Rahul and Priya jog around a circular track. Rahul takes 16 minutes and Priya takes 20 minutes for one round. They start together from the same point at 6:00 a.m. in the same direction. When will they next meet at the starting point? 3

OR

Find the smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Using the Fundamental Theorem of Arithmetic, find the HCF and LCM of 144, 180 and 192. 5
(ii) Show that, for these three numbers, HCF $\times$ LCM $\neq$ product of the numbers.

OR

Prove that $\sqrt{7}$ is irrational. Hence prove that $2 + 3\sqrt{7}$ is irrational. 5

20. **Case study:** The floor of a room measures 8 m 96 cm by 6 m 72 cm. It is to be paved with identical square tiles. 5
(i) Find the side of the largest square tile that can be used.
(ii) Find the least number of such tiles required.
(iii) If tiles of side 32 cm are used instead, how many tiles are needed?

OR

Case study: Traffic lights at three road crossings change after every 48 s, 72 s and 108 s respectively. They change simultaneously at 7:00 a.m. 5
(i) Write the prime factorisation of 48, 72 and 108.
(ii) At what time will they next change simultaneously?
(iii) How many times will they change together between 7:00 a.m. and 8:00 a.m. (after 7:00 a.m.)?

END OF QUESTION PAPER — Total: 40 Marks